

Examiner reports

Q1.

- (a) Very well answered by the majority of candidates. Most candidates used the correct identity and were successful in answering this part of the question.
- (b) Most candidates were successful in answering this part of the question. When answered correctly the factorisation of the resulting quadratic or use of the quadratic formula was usually well done but there were some sign errors. Many totally correct solutions were seen.

Q2.

- (a) This was well answered by the majority of candidates with many fully correct responses. The main error with the stretch was using the $\wedge$ -direction with a scale factor of 2 and with the translation the vector often contained $+1$.
- (b) This part was well answered by many candidates.
- (c)
 - (i) This was not very well answered by weaker candidates with many just rearranging the expression and then writing down the answer given without any convincing step being shown.
 - (ii) This was answered better than part c(i) with many correct factorisations seen. The main error was with the rejection of the solution $e^{2x} = -1$.
- (d) Some candidates obtained the correct answer and earned full marks. Most candidates made a reasonable attempt at the integrals with reasonable success and most showed enough to earn the B mark for subtraction of their areas. Incorrect limits of 6 and 0 often spoiled what would have been very good attempts.

Q3.

In part (a)(i) the completion of the square was done successfully by most candidates, although occasionally the value 6 was seen instead of 4 for q .

In part (a)(ii) it was necessary to comment on both parts of the expression; $(x + 1)^2 \geq 0$ and hence adding 4 implies that $(x + 1)^2 + 4 > 0$ for all values of x . Because of the word “hence”, an argument based on algebra rather than the features of a curve was required; for instance, an answer explaining that $(x + 1)^2 + 4$ has a minimum value of 4 was acceptable, but a statement about the curve having a minimum point at $(-1, 4)$ was not.

In part (b)(i) most candidates were able to write down the correct minimum point. A few chose to use differentiation but sometimes made arithmetic slips in finding the coordinates of the stationary point.

In part (b)(ii) those with the correct minimum point were usually able to produce a correct sketch, although the value of the y -intercept was sometimes missing. Some credit was given to candidates with an incorrect minimum point, usually $(1, 4)$, provided their graph was consistent with this minimum point.

The more able candidates earned full marks in part (c). The term **translation** was

required but generally the wrong word was used or it was accompanied by another

transformation such as a stretch. A very common (but incorrect) vector stated was $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

Q4.

Part (a) was well answered, with most candidates gaining both marks. The most common

error was $3 \ln e^x = \ln 4$, resulting in $x = \frac{\ln 4}{3}$.

Part (b) was generally well done.

Part (c) was not very well answered. Most candidates obtained 1 mark for the y -values but many candidates stopped at this stage and therefore lost the final 2 marks. The error in part (a) was often repeated.

Q5.

Completing the square was done well by most candidates in part (a)(i), although quite a few wrote q as 17 instead of 1.

Part (a)(ii) of this question was answered very badly with many giving their answer as coordinates. Candidates were either “hedging their bets” or were simply presenting the coordinates of a minimum point of a curve as their answer.

In part (a)(iii) many candidates obtained the correct value for x in this part, but there was confusion with many about how to answer parts (i) and (ii). The question was deliberately designed to test the understanding of the minimum value of a quadratic expression and when this occurred. Those who wrote “(ii) 4 and (iii) 1” scored no marks at all for these two parts of the question.

In part (b)(i) practically everyone scored a mark for multiplying out $(x - 5)^2$ correctly.

In part (b)(ii) only the best candidates obtained a correct expression for AB^2 and then completed the resulting algebra to obtain the printed answer.

It was good to see that many saw the link between the various parts in part (b)(iii). Many more able candidates substituted $x = 4$ into the expression and obtained $AB^2 = 2$, but they then failed to take the positive square root in order to find the minimum distance.

Q6.

Most candidates scored the mark for the correct printed equation in part (a), but some omitted “= 0” and others made algebraic slips when taking terms from one side of their equation to the other.

In part (b)(i) only the more able candidates were able to obtain the printed inequality using correct algebraic steps. Many began by stating that the discriminant was less than 0, clearly being influenced by the answer. Not all assigned the correct terms to a , b and c in the expression $b^2 - 4ac$ and others made sign errors when removing brackets.

The factorisation was usually correct in part (b)(ii), but many wrote down one of the critical values as $\frac{1}{3}$. Most found critical values and either stopped or immediately tried to write down a solution without any working. Candidates are strongly advised to use a sign diagram or a sketch showing their critical values when solving a quadratic inequality.

Q7.

This question proved to be the downfall for many of the candidates. It was the poorest answered question on the paper.

Many candidates answered part (a) correctly with the correct answer often seen. A common error was not to give the answer in an exact form, but this only affected candidates who showed no working. The solution $x = \ln(5/2)$ was also common.

In part (b) the only candidates who were really successful here were the ones who used a substitution such as $y = \ln x$, who then were able to formulate and solve the quadratic very easily. Those candidates who attempted to work in $\ln x$ usually (despite condoning poor notation) failed to obtain a quadratic and hence scored zero. Many candidates just substituted $2\ln x = 5$ and verified that it fitted the equation.

Q8.

In part (a), it was only necessary to complete the square for the y -terms. As a result, there were probably fewer errors this year expressing the left-hand side of the equation of the circle as $(y - 5)^2$. However, the right hand side was often written as $\sqrt{5}$, -5 or -45 instead of 5.

In part (b), quite a number who had the correct circle equation in part (a) wrote the coordinates of the centre as $(5, 0)$ or $(0, -5)$. Generous follow through marks were awarded for the radius provided the right-hand side of the equation had a positive value.

The wording in the question reassured most, though, that the radius was $\sqrt{5}$.

In part (c)(i), those with poor algebraic skills, often writing $2x^2$ instead of $(2x)^2$, struggled to establish the given quadratic equation. Also, quite a few made errors in their working but miraculously wrote down the given equation on their final line. A surprising number derived an equation in y . Quite a few simply solved the given quadratic equation in this part and thus failed to show an understanding of what was required.

In part (c)(ii), it was necessary to state that the equation had a repeated root of $x = 2$, or to use the zero value of the discriminant to show that the equation had equal roots, and hence to conclude that the line was a tangent to the circle.

In part (d), far too many simply substituted the coordinates of the point Q into the equation of the circle obtaining a nonsensical statement such as " $-3 = 0$ so the point lies inside the circle". It was necessary to see that the distance CQ was being calculated and then concluded that this distance was less than the radius of the circle, and hence the point Q must lie inside the circle.

Q9.

Candidates did not seem confident working with a quadratic expression where the coefficient of x^2 was negative. Throughout this question, candidates chose instead to work with the expression $x^2 + 8x - 9$, or the equation $x^2 + 8x - 9 = 0$, and lost quite a lot of marks.

In part (a), a large number of candidates could not factorise the given quadratic correctly, a few clearly not even recognising what was required.

In part (b), those who kept brackets in their working were usually successful in proving the identity. Some able candidates started with $9 - 8x - x^2$ and showed their skill in completing the square.

In part (c), quite a large number of candidates seemed unfamiliar with the terms “line of symmetry” and “vertex” and certainly failed to see the link with part (b) of the question. Some stated that the coordinates of the maximum point were $(-4, 25)$ and then wrote the coordinates of the vertex as something entirely different.

The sketches were somewhat varied: some found the wrong x -intercepts and drew a curve through these points; those who had completely changed the question into $y = x^2 + 8x - 9$ had a U-shaped graph. Those who drew a graph with the vertex in the correct position and with the correct shape usually had the y -intercept marked correctly as 9. However some drew their curve with a maximum point on the y -axis.

Q10.

In part (a), some weaker candidates did not realise how to derive the given equation, and others made algebraic slips when proving the printed result, or failed to write “= 0”.

In part (b), the condition for two distinct points of intersection required candidates to use the condition that $b^2 - 4ac > 0$ at any early stage of their argument. Those who simply wrote “> 0” on their final line of working, without any previous reference to the discriminant being positive, failed to convince the examiners that they deserved full marks.

In part (c), quite a number were unable to factorise the quadratic correctly and many resorted to using the quadratic equation formula to find the critical values. Where this was done correctly but left in surd form, it was given due credit except for the final mark. Very able candidates can write down the answer to the inequality once they have factorised the quadratic but far too many guessed at answers and an approach using a sign diagram or sketch is recommended. Candidates also need to realise that the final form of the answer

cannot be written as $\frac{14}{9} < k < -2$.

Q11.

In part (a), many candidates gave the correct value for where the curve met the y -axis, but a significant minority drew no part of the graph in the second quadrant, their graphs stopping at the point $(0, 1)$. Many candidates scored at least half marks for their descriptions of the geometrical transformations in part (b). The most common wrong answer in part (b)(i) was “a stretch of scale factor 2 parallel to the y -axis”, and in part (b)(ii) the direction of the translation vector caused the most problems.

Part (c)(i) was poorly answered, with many candidates failing to give sufficient justification for the printed answer. A very common error was to write 9^x as 3^y and yet still reach the correct factorised quadratic equation.

Although part (c)(ii) was answered better, the majority of candidates failed to gain all the four marks. The most common wrong approach was to take logarithms of each term in the given equation. Those candidates applying this incorrect method had clearly not seen the implication of the “Hence”. Those candidates who had seen the link and started with $3^x = 1$ and $3^x = 2$ generally scored all four marks.

Q12.

In part (a), there were some very good sketches, although not all indicated the values of the intercepts as requested. The most common omission was the x -intercept of the line, $(\frac{1}{3}, 0)$, possibly because it involved a fraction. The other common error was the omission of the y -intercept of the curve. Many wrongly assumed $(0, -3)$ to be the minimum point of the curve. A few drew the parabola upside down but were given some credit if they marked the correct intercepts on the x -axis. Many who plotted points produced a J-shape that failed to cross the x -axis twice.

In part (b), there were many satisfactory solutions, although quite a few candidates presented their proof in part (a) or (c) of the question. Some omitted “= 0” even though the equation was printed in the question paper. A few candidates guessed two values, possibly from their graph, and merely checked them in the given equation and scored no marks. Many weaker candidates made no attempt at this part.

Part (c) was answered well, although a few stopped after finding only the x -coordinates. Some wrongly substituted these values into $x^2 - x - 2$, instead of the equation of the line or curve, and thus failed to find the y -coordinates of the points of intersection. Quite a few picked out at least one of the pairs of coordinates, possibly using their graphs, and this was given credit.

Q13.

This question was not answered very well in general. The fractional values of p and q caused problems.

In part (a), although many found the value of p correctly, far fewer obtained the correct value of q . Once more arithmetic errors occurred, particularly when calculating $4 - \frac{9}{4}$.

Very few candidates understood what was required for part (b). Many hedged their bets and wrote their answer in coordinate form, $(1.5, 1.75)$, whilst many others thought the minimum value was 1.5.

In part (c), many earned a mark for the word “translation”, provided it was not combined with another transformation. However, the wrong vector was very common; few recognised the link with part (a) and used 3 and 4 instead of their values of p and q .

Q14.

The most common error by far in part (a) was the omission of brackets in $(4k)^2$. Usually $16k^2$ appeared after $4k^2$ had been seen in the line above. Candidates must be far more careful when proving a given answer in order to convince the examiner. It is essential to address the inequality by stating the condition for real roots early in the proof, rather than just inserting it into the final line.

In part (b), factorisation of the quadratic proved too difficult for many, with the most common error being wrong signs in the brackets or the roots. Some used the quadratic formula, but the calculation of $(-9)^2 - 4 \times 4(-9)$ defeated many.

Of those who calculated the critical values correctly, many assumed that they had answered the question and stopped. It was good to see more candidates showing their method of solution of the inequality by drawing a sketch or sign diagram. It was a pity that many of those who obtained the correct answer then wrote their final answer as $3 \leq k \leq$

$$-\frac{3}{4}.$$

Q15.

Part (a)(i) It was disappointing to see many candidates unable to rearrange $3x + 5y = 8$ to make y the subject in order to find the gradient. Some were successful in finding a second point on the line such as $(1, 1)$ and then using the coordinates of A to find the gradient of AB .

Part (a)(ii) Most candidates knew how to find the gradient of a perpendicular line, but those using $y = mx + c$ made more arithmetic slips than those using the more appropriate form

$$y - y_1 = m(x - x_1).$$

Part (b) Apart from those who used the wrong pair of equations, this part was usually answered correctly.

Part (c) Although this part was meant to be challenging, there were many successful attempts, particularly by those who used a sketch and reasoned on a 3, 4, 5 triangle. It had been intended that candidates would have formed an equation such as $16 + (k + 2)^2 = 25$, but more commonly something such as $16 + y^2 = 25$ was seen, resulting in the incorrect values of ± 3 .

Q16.

Part (a) The condition for real roots was not widely known and the form of the printed answer caused many to write the condition as $b^2 - 4ac \leq 0$.

Part (b) It was disappointing to see many unable to factorise the quadratic correctly. Far too many guessed at answers and an approach using a sign diagram or sketch is recommended. Candidates also need to realise that the final form of the answer cannot be written as " $k \geq -5$ or $k \leq 8$ "

Q17.

Most candidates just wrote down the required expression for $\cos^2 x$, although some worked it out from $\cos^2 x - \sin^2 x$ or $\cos(A + B)$. Omission of the 2 was a relatively rare error. Most candidates were able to derive the result in part (b)(i) although some were penalised for being too brief. Most candidates too were able to solve the resulting quadratic equation, with some making it a little more difficult than intended by reading the 1 on the right hand side as 0. Most candidates found two results for angle x in the required range. Part (c) was less successful: although most candidates understood that they needed the integral in terms of $\cos^2 x$, there were many algebraic slips, and sign and coefficient errors in the integral of $\cos^2 x$ itself.

Q18.

Part (a)(i) The completion of the square was done successfully by most candidates, although occasionally $+44$ was seen instead of -6 for q .

Part (a)(ii) Most candidates were able to write down the correct minimum point, although some wrote $(5, -6)$ as the vertex. A few chose to use differentiation but often made arithmetic slips in finding the coordinates of the stationary point.

Part (a)(iii) Although there were correct answers for the equation, the term “line of symmetry” was not well understood by many; typical wrong answers were $y = -6$, the y -axis and even $y = -x^2 + 10x + 19$ or other quadratic curves.

Part (a)(iv) The more able candidates earned full marks here. The term **translation** was required but generally the wrong word was used or it was accompanied by another transformation such as a stretch. The most common (but incorrect) vector stated was

$$\begin{bmatrix} 10 \\ 19 \end{bmatrix}.$$

Part (b) There were a number of complete correct solutions here. The errors that did occur usually stemmed from sign slips in rearranging the equations. Some candidates found the x -coordinates and made no attempt at the y -coordinates. A few candidates wrote down the coordinates of at least one point without any working.

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Q19.

Part (a)(i) Most candidates realised the need to find the value of $f(x)$ when $x = 1$. However, it was also necessary, after showing that $f(1) = 0$, to write a statement that the zero value implied that $x - 1$ was a factor.

Part (a)(ii) Those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

Part (a)(iii) This section seemed unclear to some candidates. Many tried to find the discriminant but used the coefficients of the cubic equation. Many who used the quadratic thought that in order to have one real root the discriminant had to be zero, no doubt thinking the question was asking about equal roots. Some correctly stated that 1 was the only real root but many were obviously confused by the terms “factor” and “root” and stated that “ $x - 1$ was a root”.

Part (b)(i) Most candidates were well versed in integration and earned full marks here.

Part (b)(ii) The correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 1 and 2. Very few candidates realised the need to find the area of a triangle as well and so failed to subtract the value of the integral from the area of the triangle in order to find the area of the shaded region.

Q20.

Part (a) Only the more able candidates were able to complete this proof correctly. Many began by stating that the discriminant was less than or equal to zero, no doubt being influenced by the printed answer.

Part (b)(i) The factorisation was usually correct.

Part (b)(ii) Most candidates found the critical values, but many then either stopped or wrote down a solution to the inequality without any working. Many candidates wrongly

thought the solution was $k \leq \frac{1}{2}, k \leq 2$. Candidates are advised to draw an appropriate sketch or sign diagram so they can deduce the correct interval for the solution.

Q21.

- (a) (i) Most candidates were familiar with the idea of “completing the square” and answered this part satisfactorily. There were occasional sign errors and $+9 - 4$ was not always evaluated correctly.
- (ii) There were several correct answers although some wrote (5, -2) instead of (5, 2). Some did not recognise the link between parts (i) and (ii) and chose to differentiate instead. This was a satisfactory, though more time-consuming, alternative method. Some earned no marks here as they wrote comments such as “5 is the minimum”, with no link to the y -coordinate being 5.
- (b) (i) This simple proof was usually well done. Occasionally the mark was lost due to the omission of “= 0”.
- (ii) Many scored full marks here. Most factorised the equation and obtained the correct x -values. Some made no further progress, while a few substituted into the given quadratic equation and obtained $y = 0$, instead of using the equation of the line or curve to find the values of y . It was encouraging to see many factorising the quadratic correctly. Those who used the quadratic equation formula or completion of the square often made more errors than those who factorised.

Q22.

- (a) There were several completely correct proofs here. Some lost the last mark by concentrating on the discriminant but failing to equate it to zero. There was a little fudging by some; for example, some who wrote $-4(4m + 1) = -16m + 4$ still managed to obtain the correct printed equation.

Some of the weaker candidates found $b^2 - 4ac$ using numerical values from the equation they were supposed to establish.

- (b) Almost all candidates found both values of m successfully. A few spotted just one answer and some factorised correctly and then wrote $m = -2$, $m = -6$, but they were in the minority.

Q23.

Although this question on logarithms was answered better than corresponding questions in summer 2005, the topic continues to be one which candidates find difficult. Many candidates gained credit for writing ' $2\log n = \log n^2$ ' but then often made the error ' $\log(5n - 24) = \log 5n - \log 24$ '. In part (b) it was disappointing to find so many candidates unable to solve the quadratic equation. It was not uncommon to find solutions which involved the wrong factors of the form $(n \pm 24)(n \pm 4)$. In general, only the better candidates gave a fully correct solution for both parts of this question.

Q24.

Part (a) Many candidates began by finding the correct values of p and q . A few wrote $(x - 4)^2$ and some added 16 instead of subtracting 16 so $q = 35$ was sometimes seen.

Part (b) Very few chose to consider the expression they had in part (a). Practically all candidates decided to find the discriminant instead but its evaluation was often incorrect. Not everyone quoted the expression for the discriminant, $b^2 - 4ac$, correctly. Some attempted to refer to the fact that the curve was completely above the x -axis but did not, in general, complete their argument.

Part (c) The graphs here were disappointing. Although most drew a quadratic shape, there seemed to be little reference to their part (a) and many just tried to plot a few points. Most were able to state the intercept on the y -axis. However, sometimes the point (0.19) was shown as the minimum point or a straight line intercept. Several curves were drawn only in the first quadrant, regardless of whether the quoted minimum point was (-4.3) or (4.3) .

Part (d) This was not well answered. The term **translation** was required but generally the wrong word was used or it was accompanied by another transformation such as a

stretch. The most common incorrect vector stated was $\begin{bmatrix} 8 \\ -19 \end{bmatrix}$.

Q25.

Part (a)(i) It was apparent that some candidates had not covered this part of the specification and they made no progress. Most who had done so, earned the marks here. However a few wrote $(x + 2)^2$ or even $(x - 2)^2$ and then gave the centre as $(-2, 0)$ Some managed to incorporate the 7 with the y term so wrote the coordinates of the centre as $(2, -7)$.

Part (a)(ii) Most candidates were successful in finding the correct radius. However some forfeited one mark by 'meddling' with their equation and putting 18^2 or $\sqrt{18}$ on the right hand side of the equation.

Part(b) This part was rarely attempted. Even where a correct diagram was drawn, few recognised that the chord would be bisected. Many assumed that the triangle was

right-angled at the centre of the circle. Others drew tangents instead of a chord.

Part (c)(i) Many made little progress here. However, it was good to see more able candidates coping well. A few fell at the final line writing $(k - 1)$ instead of $(k + 1)$; a few lost a mark by not introducing ' $= 0$ ' as part of the equation of the circle and simply added ' $= 0$ ' at the end of several lines of working so as to match the printed answer. Many made a slip in squaring $(2k - x)$ and some made gross errors such as writing this as $4k^2 + x^2$ or $4k^2 - x^2$. Others 'simplified' the equation to $(x - 2) + (2k - x) = \sqrt{18}$.

Part (c)(ii) Those candidates who made progress here needed both knowledge and algebraic skills and only a small minority completed this part correctly. However more earned some method marks. Use of the correct condition on the discriminant was required but some just tried to solve the equation using the quadratic formula or used ' > 0 ' instead of ' $= 0$ '. A few attempts at completing the square were seen but most failed to equate the expression to zero.

Part (c)(iii) Many candidates who had made no progress in the rest of the question stated that the line would be a tangent to the circle; however several candidates wrote at length about various transformations and completely missed the point.

Q26.

- (a) It was disappointing to see a number of candidates unable to deal with the minus sign outside the second bracket. Another very common error was to simplify the expanded brackets correctly to $-11k^2 - 14k + 25$ and then to say that this was equal to $11k^2 + 14k - 25$, thus losing the accuracy mark.
- (b)
 - (i) The sign in the printed answer caused many to write the condition for real roots as $b^2 - 4ac \leq 0$. There was a lot of 'fudging' in this part of the question and it was a rare event to see a candidate score full marks.
 - (ii) The most successful candidates identified the critical values and then used a sign diagram or a sketch to identify the correct solution to the inequality.

Q27.

The variation between candidates was immense. Some did not appear to have covered this topic in any depth.

Part (a) About two thirds of the candidates found the correct values of p and q but it was alarming to see $(x - 3)^2 - 9 + 16$ become $(x - 3)^2 + 5$ or $(x - 3)^2 + 25$ on so many occasions.

Part (b)(i) Only the confident candidates wrote down $(3, 7)$ for the vertex with no further working. Many used differentiation and found the correct coordinates, but others simply wrote either 3 or 7 as their answer. Quite a few felt that the minimum point was $(-3, 7)$. Common wrong answers from erroneous thinking were $(3, 0)$, $(7, 0)$, $(0, 16)$ or $(7, 16)$.

Part (b)(ii) Almost everyone drew a parabola though a few failed to sketch beyond the minimum point; a few graphs had a maximum point and a few cubic graphs were seen. However the positioning of the quadratic graph was crucial and often bore no connection to their vertex or the correct curve. To gain full marks the parabola had to be correct with

the y intercept value of 16 clearly indicated.

Part (b)(iii) Correct curves were generally followed by correct equations for the line of symmetry, but a few gave their answer as $y = 3$. A significant number made no attempt at this part of the question.

Part (c) Full marks here could only be obtained by using the correct term '**translation**'. Using words such as 'shift', 'move' or 'transformation' were not good enough for full marks. The best candidates used a vector, but correct descriptions of how many units the curve had been translated in appropriated directions were of course credited. However, many included a second transformation such as a stretch and this negated the first mark.

Many candidates had incorrect vectors such as $\begin{bmatrix} -3 \\ -7 \end{bmatrix}$ or $\begin{bmatrix} 3 \\ 7 \end{bmatrix}$, or most commonly $\begin{bmatrix} 6 \\ 16 \end{bmatrix}$.

Q28.

Part (a) Practically everyone made a reasonable attempt to multiply out the brackets but the minus signs caused problems for the weaker candidates, particularly when trying to combine like terms.

Part (b) Usually candidates were able to find the single real root 2. It was expected that candidates would have found the discriminant of the quadratic and, since this was negative, conclude that the quadratic equation had no real roots. All too often, candidates tried to find " $b^2 - 4ac$ " for the cubic $x^3 - x^2 + x - 6$.

Q29.

Part (a) Many candidates wrote the statement $mx - 1 = x^2 - 5x + 3$ followed by the printed answer. An intermediate step such as $x^2 - 5x - mx + 4 = 0$ was expected since the result was a printed answer

Part (b) Very few correct solutions were seen to this part of the question. The condition for equal roots was not clearly understood by many and all sorts of algebraic errors occurred when trying to square $-(5 + m)$. A very common mistake after incorrect squaring was to write $25 + m^2 = 16$ and then final answers of ± 3 were given for m . An alternative approach was often successful. This involved a realisation that the quadratic must be $(x + 2)^2$ or $(x - 2)^2$ and hence the two possible values of m could be found.

Part (c) It was very rare to see a correct description of the line being a tangent to the curve. This might be a very useful question for future candidates to practice in order to understand the geometrical situations and algebraic relationships when a line intersects, does not intersect or is tangential to a given parabola.

Q30.

Most candidates were able to state and use the identity ' $\tan \theta = \frac{\sin \theta}{\cos \theta}$ ' to obtain the printed result in part (a)(i). The identity ' $\cos^2 \theta + \sin^2 \theta = 1$ ', required in part (ii), was less well recalled, with sign errors and ' $\cos \theta + \sin \theta = 1$ ' seen. Many candidates correctly solved the equation in part (b)(i) but the explanations given for part (b)(ii) were poor. The

most common wrong answer was 'because $\cos \theta = -4$ is negative.' Those who realised the link between the parts normally gained the marks for (b)(iii) and part (c), but many tried to solve part (c) from scratch, despite the clear instruction to 'Write down' the values.

Q31.

The great majority of the candidates were able to answer both parts of this question correctly. However, a small number of candidates lost the last accuracy mark which was available for part (b) by not showing enough working. Centres should advise their candidates to show sufficient evidence of working when the answer to a question is given on the question paper.

Q32.

The responses to part (a) were very good. The vast majority of candidates were familiar with the kinematics equations for the motion of a projectile.

Many candidates answered part (b)(i) of the question correctly. However, some candidates misunderstood the axes and substituted -1 instead of 1 for y . Evidently, because the answer was given, some of these candidates were able to rectify their mistakes but others gave wrong answers.

The candidates understood and used the condition for real solutions for the quadratic equation to arrive at the relationship requested for part (b)(ii). For part (b)(iii), almost all candidates substituted 5 for R . However, some candidates were unable to proceed further and solve the quadratic equation in u^2 in order to find the minimum speed of projection. There were other candidates who attempted to use differentiation to find the minimum speed. These candidates tried to minimize the expression $u^4 - 20u^2 - 2500$ rather than finding the minimum value of u .